

Md Shakib Mondal

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Summary

Results-driven Python Backend Engineer with 4+ years of experience designing high-throughput microservice architectures, asynchronous APIs (FastAPI, Django REST), and cloud-native solutions on AWS and Kubernetes. Proven track record of optimizing system performance at scale—achieving up to 40% throughput gains and eliminating production outages—while building enterprise-grade AI pipelines with multi-agent orchestration and RAG. Seeking to leverage deep backend and distributed systems expertise to drive engineering impact at a high-growth organization.

Experience

Infosys - Senior Python Backend Engineer

Kolkata, India

- Project Account – Apple

March 2025 – Present

- Architected the migration of 3 high-traffic Python microservices from Django REST to asynchronous FastAPI, increasing system throughput by 40% and reducing p95 latency under peak load.
- Optimized application reliability for production deployments by introducing **database connection pooling** and tuning **Kubernetes resource limits**; eliminated **OOM crashes** and maintained flat memory profiles across distributed clusters.
- Designed a highly concurrent REST backend for a data planning dashboard, engineering an event-driven pipeline to aggregate and serve SingleStore metrics to the UI with sub-second response times.
- Implemented a scalable data ingestion architecture using AWS S3, enabling parallel processing of bulk uploads and integrating asynchronous event notifications for downstream services.
- Remediated security vulnerabilities across a 20-node microservice architecture, sanitizing data inputs and refactoring legacy data access layers to eliminate SQL injection vectors.

Accenture - Python Backend Engineer

Kolkata, India

- Project Account – SAP

August 2021 – March 2025

- Designed and deployed a scalable Python/FastAPI orchestration service to automate enterprise-wide provisioning workflows, successfully processing 100K+ daily requests and reducing manual operational overhead by 40%.
- Engineered an end-to-end identity management API workflow, deprecating legacy manual support processes and seamlessly handling 1,000+ concurrent state-change requests per month.
- Developed an asynchronous data archiving pipeline processing up to 5 GB of remote telemetry logs per batch, utilizing AWS S3 for secure, fault-tolerant long-term storage and auditing.
- Containerized the application suite using Docker for VPC deployment and implemented robust exception handling protocols, reducing post-deployment regressions by 45%.

Education & Certifications

Maulana Abul Kalam Azad University of Technology, West Bengal

July 2017 – June 2021

BTech in Computer Science - DGPA: 8.96 / 10.0

Claude Certified Architect - Foundations (Badge)

August 2026 – August 2027

Anthropic

Projects

Multi-Agent AI Extraction Pipeline

 GitHub

- Designed a distributed multi-agent architecture (Google ADK, LangGraph) utilizing 3 agents and 9 external tools communicating over MCP and a custom Agent-to-Agent protocol with automatic error fallback routing.
- Integrated Azure OpenAI (GPT-4o) via a LiteLLM gateway to execute a 3-phase hybrid validation (regex + LLM) pipeline for multilingual data extraction across PDF, DOCX, and TXT formats.
- Built a hybrid-retrieval RAG engine (FAISS + BM25, 65/35 weight) structured as a 4-node computational graph (retrieve, re-rank, synthesize, evaluate) querying an ERP knowledge base.
- Implemented enterprise-grade AI guardrails including Microsoft Presidio for PII sanitization and end-to-end Langfuse distributed tracing to propagate correlation IDs across network boundaries.

Technologies

Languages: Python (asyncio, pytest), SQL, Bash

Backend & Cloud: FastAPI, Django REST Framework, SQLAlchemy, AWS S3, Docker, Kubernetes, Kafka, REST APIs, Microservices

Databases & DevOps: PostgreSQL, MySQL, SingleStore, Grafana, GitHub Actions, CI/CD Pipelines

AI/ML Architecture: LangGraph, Google ADK, RAG, FAISS, BM25, LiteLLM, Langfuse, Azure OpenAI